

Reproduction of tinygrad/tinygrad: hlb_cifar10 and beautiful_mnist examples

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Independent reproduction · Benchmark or leaderboard entry · Machine learning

Abstract

Ran two training examples from tinygrad at the pinned commit, unmodified and with their default settings, on the NV backend: examples/hlb_cifar10.py (the hlb-CIFAR10 speedrun port, 1000 steps at batch size 512, fixed seed 201 from the script hyperparameters, test accuracy evaluated by the script) and then examples/beautiful_mnist.py (small convolutional network, Adam, 70 steps at batch size 512, test accuracy at the end). Training loss every 50 steps, CIFAR-10 test accuracy and the final MNIST loss and test accuracy are parsed from the script output. CIFAR-10 and MNIST fetched by tinygrad before the sealed run; CUDA NVRTC 12.8 from conda-forge as the kernel compiler. Single NVIDIA RTX 5060 Ti 16 GB, Python 3.12.

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1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	tinygrad hlb_cifar10 + MNIST	2026-10-07	1	1	Signed by the K-Veritas server

2 Results

r1: tinygrad hlb_cifar10 + MNIST

Run	Metric	Final value	Step
1	cifar_train_loss	508.86	950
1	cifar_eval_acc_pct	93.75	1000
1	mnist_final_loss	0.08	

Run	Inline claim	Value	Phase
1	cifar_eval_acc_pct	93.75	
1	mnist_test_acc_pct	98.31	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	358.5 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	0f92772130f630c2633852c8e562be54c72249b9bccc100ce4b5fcf6e016069c	4.4 MB
r1.kvbundle.zip	bae3e78018887e89e095c6f54251e33fe7e98e443178b2ea040bb18be22af6a1	4.5 MB

5 Links and licence

Reproduction of: <https://github.com/tinygrad/tinygrad/tree/dfbdb75228cdb2ce7a7aacac803ebdd6e3db8951>

Record: CC-BY 4.0. Code: MIT, as declared by the authors. Tags: tinygrad, cifar-10, mnist, deep-learning-framework.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (tinygrad hlb_cifar10 + MNIST), kv:2610.00013v1/r1. Data hash 7a4a9afa73e75c11dd8814c9f98bd317f848d771a8272d361e229ef70aa583e7.

<https://kveritas.org/records/2610.00013v1/r1.pdf>